

The effect of low-carbon processes on industrial excess heat potentials for district heating in the EU: A GIS-based analysis

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ABSTRACT

Excess heat from industrial processes can be utilized in district heating networks, thereby reducing the primary energy demand and possibly the CO₂ emissions for generating district heating. Numerous studies found a substantial potential of industrial excess heat, but have not systematically considered future changes in the energy system that will affect its utilization potential. Industrial production is set to transform to low-carbon processes and district heating needs to be generated without the use of fossil fuels. We quantify industrial excess heat using spatial matching for the EU-27, and considering the impact of the transformation to a climate-neutral energy system. The first step identifies excess heat potentials from energy-intensive industries as point sources, and considers process changes. The subsequent step spatially matches these excess heat potentials to district heating areas using a GIS-based approach. The results show that the amount of available excess heat will decrease significantly due to industry transformation. At the same time, the utilization could be increased due to lower district heating system temperatures and expanding district heating areas, resulting in 3–36 TWh per year. At local level, industrial excess heat can make a significant contribution to the supply of district heating in the future, but the major share will need to come from renewables.

1. Introduction

The European Union (EU) has committed itself to becoming a climate-neutral society and economy, i.e. net-zero emissions by 2050. This is demonstrated in its Green Deal from 2019 to 2020 [1–3], the scenarios developed in 2020 [4], and the set of proposals in its “Fit for 55” package in 2021 to bring EU legislation in line with the climate targets [5]. In 2020, the industrial and buildings sector accounted for 34% and energy supply for 24% of GHG emissions in the EU [6]. Even though these sectors have reduced their emissions by 20% over the last decade, a fundamental transformation is needed to achieve net-zero emissions, i.e. climate-neutrality. These terms signify that a future energy system and society does not emit more GHG to the atmosphere than are removed by the planet’s natural adsorption or additional measures. About 50% of the final energy demand is used for heating, mainly for buildings and process heat in industry [7]. Thus, strategies for climate-neutral heating are essential. Strategies include the use of

CO₂-neutral energy carriers and improving energy efficiency, but also reducing the primary heat demand for district heating (DH) by using heat that is available from industrial processes, i.e. excess heat.¹

To achieve carbon-neutrality in the industrial sector, all subsectors need to undergo transformation and the following steps are crucial: First, process heat will need to be supplied by renewable energy sources [8,9]; second, process emissions stemming from chemical reactions (e.g. iron ore reduction using coal leads to CO₂ emissions in steel production) should be avoided or reduced as much as possible [9,10]; and third, material demand will change if material circularity is promoted [11]. It is expected that many industrial processes will continue to produce excess heat, which cannot be used in the process or on-site for other purposes [12–15]. However, this fundamental transformation will affect the excess heat available from industrial processes and could reduce it.

There are two main strategies to become climate-neutral in the buildings sector: increasing the insulation level of the building stock and changing the energy source for heating buildings [16]. There are

Abbreviations: BAT, best available technology; CCS, carbon capture and storage; DH, district heating; EAF, electric arc furnace; EU, European Union with 27 member states; GHG, greenhouse gases; GIS, geographic information system; H-DR, hydrogen direct reduction of iron ore; HVC, high value chemicals, olefins.

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¹ The term *waste heat* is frequently used as a synonym for *excess heat*.

national legislations in effect in the member states driven by the Energy Performance of Buildings Directive² [18], which sets energy efficiency standards, for both newly built and existing buildings. District heating can be an efficient source of climate-neutral heating [19–22], if, the provided heat is generated by climate-neutral sources. In 2020, however, 68% of DH in the EU was still based on fossil fuels [23]. Industrial excess heat could be one option for a climate-neutral DH supply.

2. Background

Several studies have quantified the possible utilization of industrial excess heat in the EU, identifying potentials ranging from 80 up to 800 TWh per year [24–27] (compared to the current DH demand of residential and non-residential buildings in 2020 of 336 TWh [28]). This identified potential could contribute to the transformation of DH generation [26,27,29–31]. These studies are based on different methods for estimating excess heat, which were described in a previous publication [27]. However, these studies did not systematically consider the transformation of the energy system and of the industrial sector to low-carbon processes. To the best of our knowledge, to date, no study assessing industrial excess heat potentials in the EU-27 has also considered the transformation of industrial production towards climate-neutral processes. This paper aims to fill this gap by matching excess heat potentials spatially to district heating areas in a climate-neutral energy system.

Spatial matching is commonly used to analyse the techno-economic potential for utilizing heat in DH systems and has been applied in national and EU-wide potential analyses, that included industrial excess heat [26,29–32]. Spatial matching considers the locations of both the heat sources and the DH systems, as well as the quantity of heat that could actually be provided and utilized in the respective areas. Several studies have pointed out that the most economic DH areas are those with the highest heat demand and thus the lowest specific distribution costs for DH [21,33–42]. However, only a few studies have included the effect of building renovation and its impact on future demand when determining economic DH areas, especially at EU level, e.g. Refs. [43–46]. In this paper, we use open data for the spatial extent of current and possible DH areas for the EU from the Pan-European Thermal Atlas (peta) 5.2 [27,47], and assume an efficiency improvement of the buildings within the identified areas.

In the future, district heating systems can have temperatures below 50 °C, often called 4th generation DH [22], which increases the possibility to utilize excess heat and renewables more efficiently, while reducing heat losses [48,49]. In a previous analysis, we found that decreasing the temperature at which the industrial excess heat is recovered from 95 °C to 55 °C increases the potentials by about 20% [27]. However, this did not take the future transformation of industry into account. Thus, we now aim to quantify the effect of a reduced system temperature when matching industrial locations with future excess heat potentials to DH areas.

This paper quantifies the potential contribution of industrial excess heat in DH grids in a future climate-neutral energy system for the EU. We particularly aim to understand how the transformation to low-carbon industrial processes will affect the excess heat available that can be used to supply DH to the building sector.

To account for the transition of the industry sector, we integrate future industrial production quantities and low-carbon processes, as the transformation of process and material demands will substantially change the available excess heat. We estimate the process-specific excess heat potentials per industrial plant and then spatially match the georeferenced available excess heat from climate-neutral industry in a Geographical Information System (GIS) to a dataset of “Current DH areas” and “Possible DH areas”. To deal with uncertainties, we construct

several scenarios and variations to show a range of possible outcomes. This analysis contributes to the discussion of the future potential of industrial excess heat for supplying DH and the impact of industrial transformation.

3. Method and data

The method consists of three steps (Fig. 1). The year 2015 is used as a status quo reference due to data availability for industrial sites and DH areas, 2050 is the year of reference for a climate-neutral energy system and the geographical reference is the EU-27.

1. *Estimation of industrial excess heat potentials*: We estimated georeferenced future excess heat potentials from energy-intensive industrial sites. The specific excess heat potential per process was based on a broad literature review. In particular, we considered future low-carbon production processes like electrified and hydrogen-based process heating and distinguished a transformation and an import scenario.
2. *Matching of DH areas and demand*: We took the two datasets available from peta 5.2 [47] and mapped the DH coverage of Current and Possible DH areas, together with the associated heat demand. We estimated the DH demand based on assumptions concerning the connection rate and improved energy efficiency of the building stock, which consequently lowered the demand for heat. We distinguished in particular system temperature levels of 25, 55 and 95 °C.
3. *Spatial matching to calculate the utilization potential*: We matched the excess heat potentials with the dataset of Current and Possible DH areas to estimate the range of potential utilization in district heating supply.

The industrial excess heat potential (based on the respective industry scenario) is combined with the Current and Possible DH areas (with different system temperatures), i.e. matched separately to these DH areas. In total, 12 combinations were calculated, showing the range of possible future developments and their influence on the results.

3.1. Estimating industrial excess heat potentials

This section describes the methodology and databases used to estimate the potential excess heat at industrial sites available for external utilization in a climate-neutral energy system. The analysis focuses on energy-intensive industries producing steel, aluminium, cement, glass, paper, chemicals and refinery products. For these processes, data on site-specific annual production are available in several databases and stored in the Industrial Database (see 2.1.2). The excess heat available at industrial sites depends on several factors that will change in the future. Important factors considered here include the future production volumes, the locations of industrial sites and the switch from mostly fossil-based to low-carbon processes.

3.1.1. Industrial production volumes and processes

The assumptions concerning future production volumes and possible technology switch to replace the carbon-intensive processes were based on the study “Industrial Innovation: Pathways to deep decarbonisation of Industry”, 2018 for DG Clima [9,50]. We used the production volumes of the scenario 4b “Mix95” with a 95% reduction of GHG emissions, as this describes an almost carbon-neutral EU industry sector with a strong role of new low-carbon processes. There are several central assumptions of the scenario that influence material demand and process design. The gross value added of all industrial sectors will grow, but energy-intensive industrial subsectors will grow at a slower rate than the other subsectors. Circular economy approaches (recycling and material efficiency) will reduce the demand for primary materials like steel or chemicals. New low-carbon processes will completely replace today’s fossil-based and emission-intensive processes. Examples include

² Negotiations are underway for a recast of the EPBD [17] as part of the “Fit for 55” package [5].

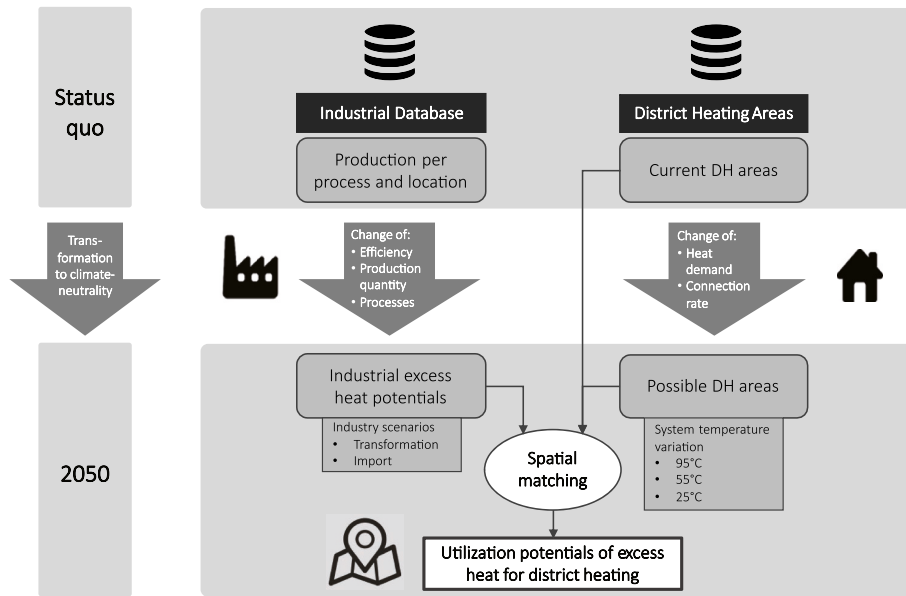


Fig. 1. Methodology applied in the study to estimate the future excess heat utilization for DH.

hydrogen-based steelmaking and electrified furnaces. CCS will be used to capture any remaining process emissions in lime and cement production. In the Mix95 scenario, electricity is the most important energy carrier, with hydrogen for high-temperature processes and as a feed-stock in chemical processes. FORECAST, which is the bottom-up industry sector model used, provides a high level of technological detail [51], which makes a process-specific estimation of excess heat potentials possible.

In addition to the production volumes at national level and the general technological pathways described above, we further assumed that energy efficiency potentials and internal excess heat recovery in all plants are exploited to the technical limits. This is a normative approach that prioritizes energy efficiency and internal excess heat use over the external use of excess heat. Furthermore, there is still substantial uncertainty about how process heat will be supplied in the future. While the DG Clima study [9] assumes widespread electrification in process heating, hydrogen is also a feasible and frequently discussed alternative for high-temperature processes [52]. Additionally, the import of basic materials could move outside the EU to locations with greater availability of renewable energy resources. Two possible scenarios are calculated in order to cope with these uncertainties:

- Transformation scenario: Industrial furnaces will be electrified where deemed feasible, based on the Mix95 scenario in Ref. [9], with the difference that hydrogen will be used in glass and cement production [52].
- Import scenario: The production output of industry will decrease as the energy-intensive basic materials ammonia, ethylene and iron sponge will be imported to the EU. Everything else remains the same as in the transformation scenario.

Excess heat from electrolyzers on site was not considered in this study. While there might be certain industrial sites that will produce hydrogen locally, this is very unlikely for large industrial plants due to enormous hydrogen demands and the need to store it locally.

3.1.2. Locations of industrial sites

The location of all the major energy-intensive industrial sites in the EU today are listed in the Fraunhofer ISI Industrial Database, which includes information about industrial production sites, their locations and associated processes together with annual production output from

emission and sectoral databases, as described in previous publications [27,53]. The sites listed in the database represent 62% of the EU's total industrial energy demand.³ These industrial sites were published together with the estimated excess heat potentials for current processes as open data in Hotmaps [45] and peta5.2 [47]. In this analysis, we assumed that the current locations will remain and all sites with similar processes will undergo the same transformation to low-carbon processes. The production volumes for 2050 at national level from the Mix95 scenario were distributed to the individual sites based on current production capacities. In the import scenario, it was assumed that the respective production plants will be shut down.

Table 1 shows the current and future production volumes, number of georeferenced installations and assumed process development for each energy-intensive product considered. For example, direct reduction of iron ore with hydrogen (H-DR) was assumed for steel production in the transformation scenario. This means that all current blast furnaces will be replaced by hydrogen shaft furnaces together with electric arc furnaces (EAF). Each of these installations will have about a third of the current capacity in the future, as overall production of primary steel will decrease. Coke ovens and basic oxygen furnaces will be phased out together with the blast furnaces and will not be replaced. In the import scenario, only EAFs were assumed at these sites, since iron sponge is imported.

3.1.3. Excess heat factors of low-carbon processes

This section describes the approach used to calculate the excess heat potential per process, presents the data used per subsector and summarizes the resulting indicators. We calculated the specific excess heat potentials per process according to the method presented in Ref. [27], which was based on [55], where applicable. We used the excess heat factors from our previous publication [27] for the current processes. The excess heat potentials for the low-carbon processes were calculated for

³ The listed sites cover almost all of the processes needing high temperatures above 500 °C and most processes with temperatures between 200 and 500 °C. Excess heat from other industrial processes was not quantified using the bottom-up methodology due to the lack of site-specific production data. These sites could contribute a smaller share of excess heat in the future, especially for low-temperature DH (missing processes with significant demand for process heat above 200 °C are gypsum, lime burning, brick, carbon black, bread and bakery, soda ash) [54].

Table 1

Current status and future transformation of processes with excess heat. Technology data and production from Ref. [9], hydrogen processes based on [52], number of sites based on [27].

Product	Annual production in Mt in EU		No. installations in EU, 2015	Process 2015	Process 2050	
	2015	2050			Transformation scenario	Import scenario
Steel, primary	92	33	52	Coke ovens	H-DR + EAF	Phased out
			56	Blast furnaces/Direct reduction natural gas		Import of iron sponge, EAF
			32	Basic oxygen furnace		Phased out
Steel, secondary	63	112	186	Electric arc furnace	Increase of secondary steel (~77%)	
Aluminium, primary	2	1	16	Bayer & Hall-Héroult process	Process 2015 remains	
Aluminium, secondary	3	3	154	Melting furnace	Increase of secondary aluminium (~70%)	
Container glass	20	18	165	Recuperative/Regenerative	Hydrogen melting	
			9	Oxy-fuel		
Flat glass	12	14	61	Recuperative/Regenerative	Hydrogen melting	
			3	Oxy-fuel		
Clinker	123	98	28	Wet	Hydrogen furnaces	
			156	Dry		
			24	Dry and preheating		
Pulp	26	18	123	Chemical pulping	Electrification	
	8	5	58	Mechanical pulping	Process 2015 remains	
	45	66	457	Recovered fibres	Electrification	
Paper	89	97	922	Paper making	Electricity for process heat and heat pumps	
Ammonia	18	14	26	Haber-Bosch	Feedstock H2	Phased out
Chlorine	11	10	9	Diaphragm/Mercury	Replaced by membrane process	
			60	Membrane	Process 2015 remains	
Ethylene and Olefins	19	20	98	Steam cracker (Refineries)	Synthesis of methanol and olefins	Phased out

three different temperature levels, at which the heat will be recovered externally: 25 °C as a lower limit, 55 °C, and 95 °C as the upper limit. We assumed that internal heat utilization potentials will be exploited with priority in the future when new processes are installed and the pressure for efficient heat utilisation increases due to higher energy costs. For

hydrogen processes, the estimation was based on a combustion calculation considering temperatures, energy carrier composition and mass flow of the exhaust gases. It calculates the enthalpy in the exhaust gases at a certain temperature based on the gas composition, which is mainly water vapour. For electrical furnaces, there is no exhaust gas from fuel

Table 2

Estimated specific excess heat factors for processes in a carbon-neutral energy system.

Product	Process technology 2050	Exhaust gas temperature (°C)	Excess Heat 25 °C (GJ/t)	Excess Heat 55 °C (GJ/t)	Excess Heat 95 °C (GJ/t)	Calculation based on	Calculation method
Steel, primary	Hydrogen direct reduction shaft furnace	250	0.19	0.04	0.03	Bhaskar et al. 2020 [56]	Exhaust gas composition based on Manz et al. 2021 [27]
Container glass	Electric arc furnace	200	0.03	0.02	0.02	Manz et al. 2021 [27]	Directly taken from source
	Electric furnace	200	0.26	0.09	0.07	Stormont 2010 [57], Ireson et al. 2019 [58]	Estimation of process efficiency
	Hydrogen furnace	200	0.37	0.13	0.09	Neuwirth et al. 2022 [52]	Exhaust gas composition based on Manz et al. 2021 [27]
Flat glass	Electric furnace	200	0.43	0.15	0.11	Stormont 2010 [57]	Estimation of process efficiency
	Hydrogen furnace	200	0.62	0.22	0.16	Neuwirth et al. 2022 [52]	Exhaust gas composition based on Manz et al. 2021 [27]
Cement clinker	Electric furnace	250	0.45	0.32	0.24	Antunes et al. 2021 [59]	Estimation of process efficiency
	Hydrogen furnace	250	0.54	0.38	0.32	Neuwirth et al. 2022 [52]	Exhaust gas composition based on Manz et al. 2021 [27]
Ammonia	Haber-Bosch with hydrogen as input	150	2.15	1.51	1.25	Smith et al. 2020 [60]	Thermodynamic analysis of process
Ethylene/Olefins	Methanol synthesis and MtO		3.80	0.97	0.56	Thonemann et al. 2019 [61] Fritz et al. 2020 [62]	Thermodynamic analysis of process
Steel rolling/forming	No fundamental changes		0.01	0.01	0.01	Manz et al. 2021 [27]	Directly taken from source
Aluminium, primary	No fundamental changes	700	1.09	1.05	1.00	Manz et al. 2021 [27]	Directly taken from source
Aluminium, secondary	No fundamental changes	538	1.09	0.71	0.65	Manz et al. 2021 [27]	Directly taken from source
Chlorine, membrane	No fundamental changes	149	0.18	0.04	0.03	Manz et al. 2021 [27]	Directly taken from source

combustion, so the excess heat factors were based on the energy efficiency values of the process. There is a lack of data for some of these processes, as they are not yet in industrial operation, although there may be larger demonstration plants. Therefore, parameters based on thermodynamic models and estimates of efficiency were taken into account, and data gaps were closed by comparisons with similar processes. These are discussed in the following paragraph on a process level. For all processes, the maximum level of internal heat recovery was assumed. The resulting specific excess heat factors for each process are summarized in Table 2.

To calculate the total available excess heat per process at one site, the annual production per process $m_{\text{product,year}}$ in tonne was multiplied with the specific excess heat factor $\frac{E_{\text{ex}}}{m_{\text{product}}}$ in GJ/t:

$$E_{\text{ex,year}} = \frac{E_{\text{ex}}}{m_{\text{product}}} \times m_{\text{product,year}} \quad (1)$$

Steel. Hydrogen-based steel production offers the possibility of a carbon-free reduction of iron ore to iron, in contrast to the conventional blast furnace and converter route based on coke [63–67]. Several pilot projects for direct reduction in shaft furnaces began in Sweden, Germany and China in 2020 [68–74].⁴ In this paper, the excess heat estimation was based on the hydrogen direct reduction and electric arc furnace route in all scenarios.⁵ The production sites of blast furnaces will be converted into H-DR shaft furnaces and EAF. The import scenario assumes that the first production stage of reducing iron ore to iron sponge is relocated to areas outside the EU where green hydrogen can be produced at lower costs, as vast amounts of hydrogen will be needed.

The temperature in the shaft furnace was assumed to be 800–900 °C [75,76]. The exhaust gas contains water and excess hydrogen, which is separated from the water in a condenser and then reintroduced to the shaft furnace after preheating. The energy from cooling in the condenser can be used to preheat the hydrogen [76]. We used the modelling results of Bhaskar et al., 2020 [56] with a calculated exhaust gas temperature at 250 °C. Based on that, we calculated the energy of the exhaust gas to assess the specific excess heat factor. We assumed that 50% of this heat is available for utilization in DH, in order to account for technical limitations, other internal heat utilization possibilities and utilization efficiencies.

Aluminium. Primary aluminium production was assumed to remain basically the same process, mainly based on electricity, so excess heat will only change due to efficiency gains and shifts towards more secondary production, which will reduce the total primary production by 30% by 2050. The excess heat values for the Hall-Héroult-process were based on [27], with a limited potential for hydrogen replacing natural gas for process heat [52]. For secondary production, typical losses for electric melting furnaces (e.g. immersion heaters) with heat recovery were based on Brough et al., 2020 [77].

Glass. In glass production, all-electric melting furnaces are already in operation, albeit with smaller capacities than gas-fired furnaces, and for special glass types [78]. More common are electric boosters, which are co-fired furnaces for quality improvements [57,79]. The study of Zier

et al., 2021 [78] gives an overview of the decarbonisation options for glass production, and regards hydrogen-fired furnaces as well as electrified furnaces as options. Electric furnaces heat the glass melt using several electrodes; microwaves or plasma could be future alternatives. Since there is no exhaust gas present, electric furnaces have an efficiency of up to 90% [58]. As no plants exist with comparable capacities to gas-fired furnaces nor detailed models of energy losses, the electricity scenario assumes that 2% of the energy input can be utilized for external use, with most of the losses being radiant losses from the furnace's walls. Hydrogen-fired furnaces require changes to the process design and future research [80,81]. For these, we used the excess heat factors from our previous study of gas-fired furnaces [27], assuming that natural gas will be completely replaced by hydrogen.

Cement clinker. Clinker calcination emits large amounts of CO₂, mainly from the chemical reaction. However, about one third of these emissions are from burning fuels and waste in the process. Hydrogen co-fired or electric plasma furnaces could be possibilities, but have not yet been constructed [59,82], and future research is needed for large-scale plants. There are no published data available on demand and efficiency of low-carbon processes for clinker production. The specific energy demand of electric kilns was assumed to be slightly lower than of the conventional process, with estimations about energy efficiency from Ref. [59]. For hydrogen, the specific energy demand values were taken from Ref. [52], and the excess heat factors are similar to those for the conventional process.

Pulp and paper. In general, most process heat in the pulp and paper subsector is used for drying processes and chemical pulp production with temperatures between 100 °C and 200 °C. It is common to use biomass in paper production, using residues like the black liquor from the processes themselves. At a paper production site, usually more low-temperature than high-temperature heat is required. The process steps using higher temperatures could be electrified or use hydrogen, and process heat on lower temperature levels offers huge potential for exploiting the excess heat [50]. Based on this, we assumed on-site utilization of this excess heat in drying processes using heat pumps with temperatures up to 150 °C, so that no excess heat remains for external use. This was also shown in the study by Rogers, 2018 [83].

Ammonia. Ammonia is conventionally produced using the Haber-Bosch process, with synthesis gas of hydrogen and nitrogen as an input. The hydrogen for the synthesis gas is currently based on steam reforming using natural gas [60]. In the transformation scenario, green hydrogen is used instead. Thus, we assumed that the energy losses from the Haber-Bosch process will remain the same in the future. The process itself is exothermic, and needs electricity for compression, auxiliary aggregates and air separation, depending on the process design. The heat from the reaction generates 2.7 GJ/t with no possible heat utilization on site [60]. After accounting for conversion losses, we assumed that any remaining heat can be utilized for district heating, depending on the temperature level.

Ethylene and high value chemicals (HVC). Refineries produce oil-based fuels for transport and feedstock for the chemical industry, mainly ethylene and high value chemicals (HVCs or olefins). Thonemann et al., 2019 [61], assumed in one scenario that the total production of ethylene and HVCs is converted to the Methanol-to-Olefines (MtO) route. This route synthesises methanol using hydrogen and carbon dioxide to produce long-chained chemicals like ethylene and other HVCs. We assumed that locations of refineries will remain and produce synthetic HVCs from hydrogen in the MtO route, as the infrastructure already exists for different materials (located close to harbours or rivers, access to pipelines for ethylene). In total, 0.43 t hydrogen is needed for 1 t of HVC. The energy input is 5 GJ/t for MtO and 12 GJ/t for methanol synthesis. This

⁴ One pilot plant for the direct reduction of iron ore with hydrogen in a shaft furnace was built in Sweden in 2020. In addition, the partial replacement of coke with hydrogen in the blast furnace route is currently being tested by Dillinger/Saarstahl and Thyssenkrupp Steel in Germany. Direct reduction plants with (partial) hydrogen usage are announced by the HBIS Group to be in operation in China, and were started in 2022 by SALCOS/Salzgitter AG in Germany, and in 2023 by ArcelorMittal in Germany, Liberty in Romania and H2GS in Sweden.

⁵ There has been recent progress in using electrolysis for iron making and this could potentially be more energy-efficient than the H-DR process [50].

is comparable to the value of 18 GJ/t in Ref. [62]. The methanol synthesis itself is an exothermic reaction [84], resulting in excess heat potentials.

3.3. Spatial analysis and matching algorithm

The excess heat potentials from industrial sites identified as point sources were mapped and matched to DH areas to quantify the future utilization of this excess heat for DH. This section presents the data sources for the DH demand and areas, the data processing as well as the matching method used. We considered different possible future DH coverages, using the open data repository from peta 5.2 [47]. Two datasets were used from the project report sEnergies D5.1 [85]: Current District Heating areas and Possible District Heating areas. These datasets can be downloaded as shapefiles with the corresponding heat demand in these areas in the attribute tables, and visualized and processed in a GIS software. Current DH areas represent the locations where DH networks currently exist, originating from the Halmstad University District Heating and Cooling database [86], with further boundary assumptions based on Heat Roadmap Europe data [37]. Possible DH areas represent areas with regional heat demand densities per hectare above 500 GJ per year, determined using the useful heat demand of buildings in the respective area [20,39]. Current DH areas can be seen as the lower limit of DH coverage in the EU. In contrast, Possible DH areas represent a DH market share of 45%, if all the buildings in these areas are connected in the EU in 2050. This can be seen as a very ambitious expansion of DH to include all areas with sufficient heating density (Fig. 2). This scenario is used to show the upper limit of industrial excess heat that could be utilized in the future. The initial datasets do not include a decrease in heat demands due to building renovation.

Two adaptations were made to consider efficiency improvements to the buildings stock and thus the reduced heating demand, and possible developments in the connection rate, i.e. the share of buildings that are connected within the DH area. Similar to building sector analyses, an average reduction of the heating demand to 70% of the current status was assumed for both Current and Possible DH areas, which is stated as necessary in an efficient energy system in Refs. [19,87,88]. The connection rate was assumed to be 46% for Current DH areas, which is the implicitly assumed connection rate in the Current DH area dataset,

to show the lower limit [27,34], and 100% for Possible DH areas to show the upper limit.

The regional analysis and spatial matching to the industrial excess heat sources was conducted in the software QGIS with the PyQGIS plugin [89] with a customized matching algorithm. Industrial locations with excess heat were matched to the most suitable DH areas, within a defined maximum distance. The PyQGIS plugin allows automated processing of regional data. The required inputs include the identified DH areas as polygons, and the excess heat sources as point layers. The matching algorithm consists of two steps (Fig. 3). First, the industrial sites with excess heat in the transformation and import scenarios were allocated to the Current and Possible DH area polygon datasets, limited by the maximum distance of 10 km from the excess heat source to the boundary of a DH area. This distance is used as a conservative break-even point of cost-effectiveness for connection pipes [27,90]. For this, a spatial grid was created to presort the heat sources to the DH areas. Otherwise, the distances from all industrial excess heat point sources to all DH areas would have been calculated. As a result of this first step, excess heat sources were allocated to all DH areas which do not exceed the maximum distance of 10 km, and for each DH area a prioritization factor was calculated based on the annual DH demand and distance from the excess heat source. In the second step, the prioritization factor was used to allocate each excess heat source to a single or to several suitable DH areas sequentially if there is more than one area within the 10 km, to prevent the closest but very small DH area being connected to an industrial site with large amounts of excess heat. This approach is quite similar to the one presented by Möller et al., 2019 [20], but our approach does not consider maximum transmission costs based on delivered heat, as we assumed a conservative maximum distance of 10 km that includes most of the industrial excess heat sources.

The algorithm calculates the supply potential, irrespective of how much of the heat could actually be consumed by the DH areas within the 10 km limit after the prioritization. The utilization potential was calculated subsequently, with the annual DH demand in the suitable DH areas as a limit of the supply potential.

4. Results

This chapter describes the results obtained from each methodological step. First, the evolution of industrial excess heat potentials, and second, the results of matching them to DH areas. Finally, we discuss and analyse the implications of industrial transformation for the possible utilization of industrial excess heat in DH systems. The results are depicted for the three possible system temperatures regarded: 95 °C, 55 °C and 25 °C.

4.1. Future industrial excess heat potentials in the EU

The annual industrial excess heat in a climate-neutral energy system ranges from 9.2 TWh (import scenario, 95 °C) up to 50.2 TWh (transformation scenario, 25 °C). These figures indicate the total potential for external use after internal heat utilization, e.g. for preheating and drying. The import scenario has lower excess heat potentials (37–48% of the transformation scenario, depending on the temperature level considered), as basic materials like iron and ethanol are imported, so there is no excess heat available from the production processes involved.

There is a general decrease in excess heat from industrial sites for all subsectors in a climate-neutral energy system, compared to the status quo (see Fig. 4 for the results for the temperature level of 55 °C). For the results of all temperature levels, the reader is referred to TableAnnex 1. The figure illustrates the different effects considered when quantifying the excess heat from low-carbon processes separately, starting from current production process and volumes, and average levels of efficiency and internal heat utilization. The effects are separated to identify the main influencing factors. One major effect on excess heat potentials is the increased energy efficiency of current processes (i.e. using BAT per

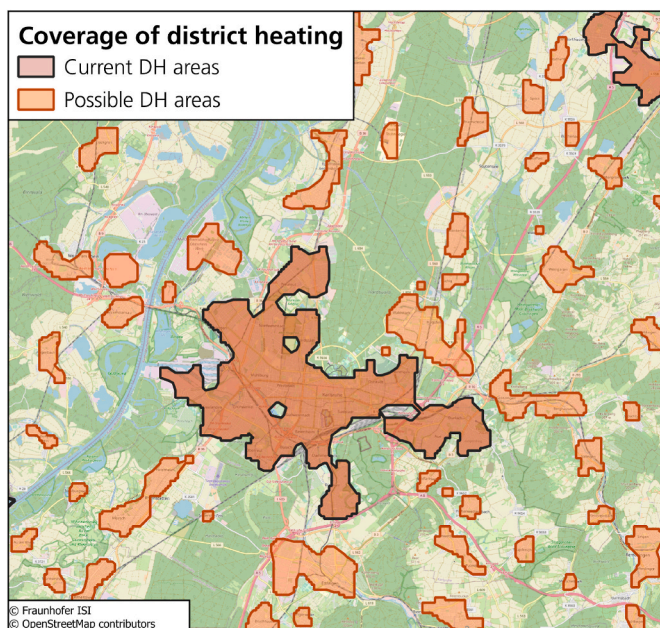


Fig. 2. Current and Possible DH areas based on the example of Karlsruhe, Germany. Own illustration based on [47].

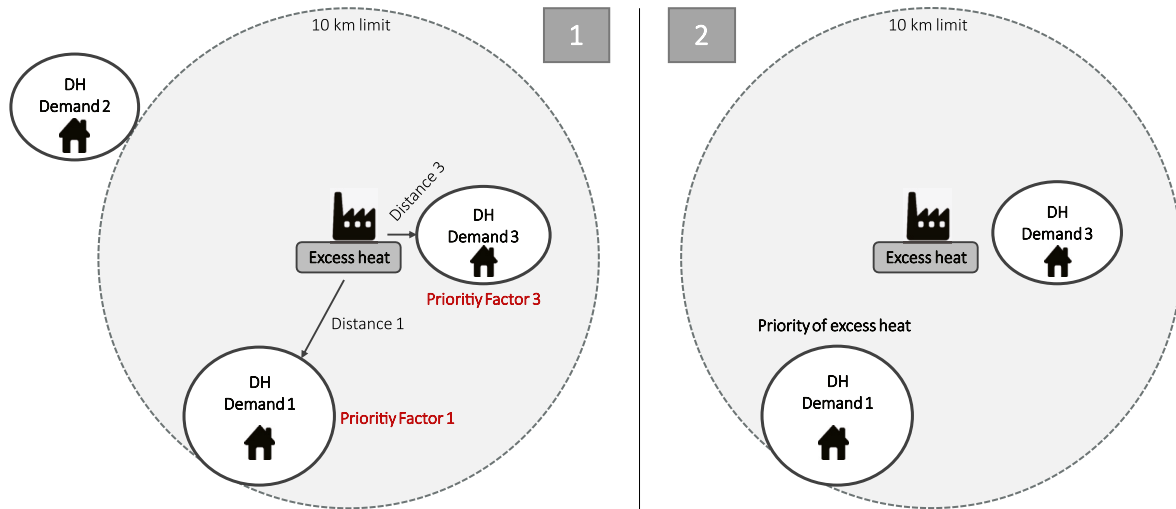


Fig. 3. Schematic overview of spatial matching algorithm. Step 1 on the left with distance calculations within a 10 km limit and step 2 on the right with prioritized sequential excess heat allocation.

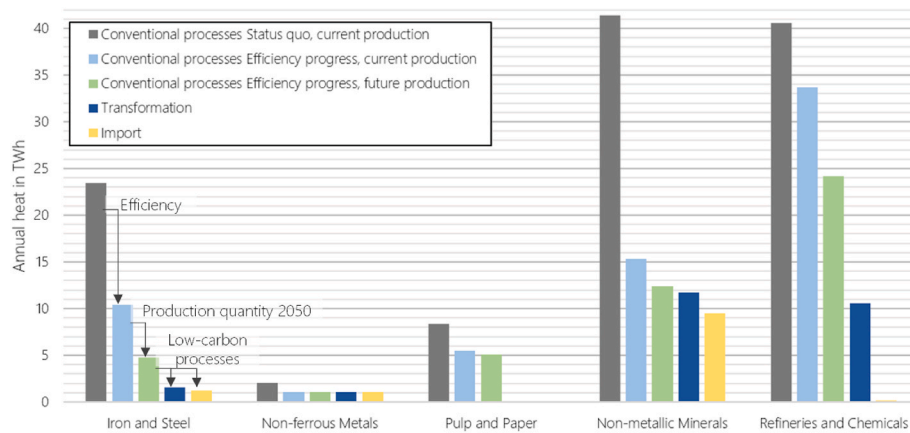


Fig. 4. Industrial excess heat potentials available per subsector in the EU for 55 °C, showing the separate effects of future developments: status quo, efficiency progress, production volumes and for the two scenarios of process transformation to low-carbon processes.

process and fully exploiting the potential of internal heat utilization). Thus, energy efficiency was incorporated using the specific excess heat factors from Ref. [27]. We first added these efficiency improvements of today's processes; second, the change in production volumes due to higher material efficiency and circularity; and third, the process transformation assumed in the transformation and import scenarios. Compared to current potentials, the future potentials decrease by 78%–80% for the transformation scenario and by 90–92% for the import scenario, depending on the assumed temperature level of heat utilization.

Efficiency progress lowers the excess heat potentials by around 40%, with the most significant decrease in the non-metallic minerals subsector. The assumed future production volumes further decrease the potentials by around 16%, compared to the status quo. This is due to a circular economy and a higher share of secondary production routes in steel, aluminium and paper as well as a lower demand for oil-based products. The implementation of low-carbon processes in the transformation scenario reduces the available excess heat potentials even further by another 20%. There is no excess heat from the pulp and paper subsector, because all excess heat is assumed to be utilized internally with heat pumps. The chemical industry and refineries have lower potentials as oil refining will be phased out in a climate-neutral energy system, with a remaining demand for petrochemicals that will be

produced synthetically. The synthesis of methanol, ethylene and ammonia could be a potential new source for excess heat. The excess heat potentials in the import scenario are 11% lower than in the transformation scenario. If basic materials are imported, there will be a substantial decrease in the potential excess heat from the chemical and refinery sites as well as from iron and steel sites decreases substantially.

The spatial distribution of industrial plants and corresponding excess heat potentials across the EU is mapped in Fig. 5 for 55 °C in the case of the transformation scenario. The map reveals industrial cluster in coastal areas or industrial regions, e.g. in the Netherlands/Belgium/North-West Germany. Few sites show high excess heat potentials above 200 GWh per year and more.

4.2. Utilization potential of excess heat for DH in the EU

The spatial matching of the industrial excess heat sources with the Current and Possible DH areas for the two industry scenarios and the different temperature levels results in utilization potentials taking into account the annual heat demand from the matched DH areas (Fig. 6). The available industrial excess heat for district heating ranges from 3 to 36 TWh per year. The value of the first bar (grey) depicts the total potential of industrial excess heat, before matching with the DH areas. Given the DH demand of 165 TWh in Current DH areas, and 1155 TWh

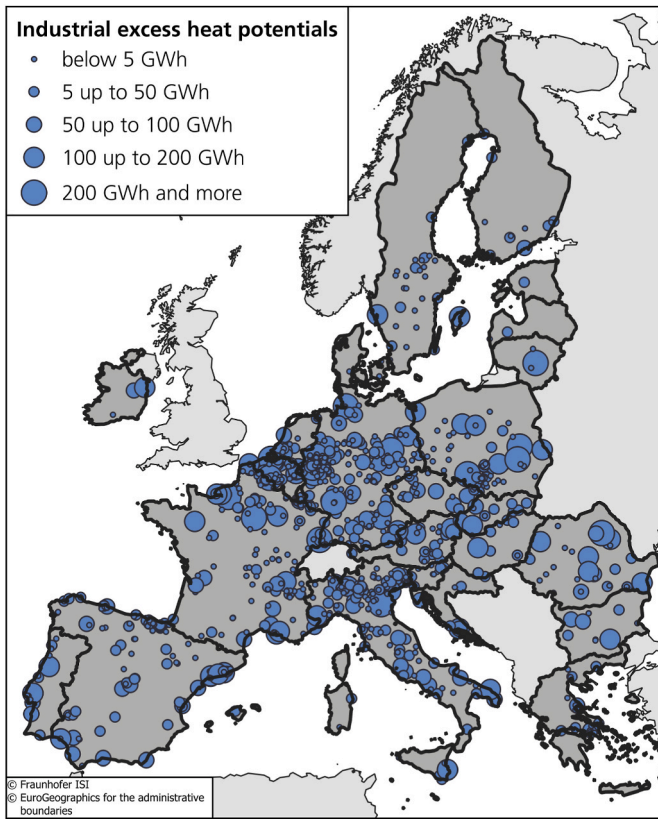


Fig. 5. Annual industrial excess heat for external use in the EU for 55 °C in the transformation scenario.

in Possible DH areas, this results in a potential share of industrial excess heat in DH of 1–8% (transformation scenario, 55 °C: 3.7%). This figure depends on the assumed DH coverage, temperature level and industry scenario. Industrial excess heat can be utilized by 316 Current DH areas, and by 885 Possible DH areas.

Two factors limit the excess heat utilization potential: the distance to suitable DH areas from an industrial site and the DH demand in the specific DH area. For all temperature levels, about 35% of the total excess heat potential can be utilized for Current DH areas, and about 75% for Possible DH areas. Thus, DH demand is a main limiting factor

(see also TableAnnex 2, where supply potentials are included). This means that the excess heat available from industrial sites is often larger than the demand for DH within 10 km. The utilization potentials in the import scenario are 31–52% lower than in the transformation scenario. Country-specific values for 55 °C in the transformation scenario are listed in TableAnnex 3.

Comparing the utilization potentials of a climate-neutral industry to the status quo shows a reduction of 63–90%, depending on temperature and DH coverage (Fig. 7). Even though the utilization potentials are substantially lower, temperature level has a strong influence and could partially compensate the reduction due to industrial transformation.

A regional analysis for the transformation scenario and the temperature level of 55 °C reveals that even though the utilization potentials make a rather small contribution to overall DH demand, in specific regions, this share could exceed 30% and even amount to 100% (Fig. 8). For illustration purposes, the DH demand and the utilization potential were aggregated for each NUTS 3 region and shaded by the share of excess heat in DH demand. There is a general decrease in the share of excess heat when comparing Possible DH areas with Current DH areas, as they have a higher demand for DH. On the other hand, Possible DH areas are more widespread than Current DH areas, thus utilizing additional excess heat from sources that are too far from Current DH areas.

At the level of individual suitable DH areas, excess heat could supply more than 90% of the demand for heat in 22% of Current DH areas and 26% of Possible DH areas. These are generally small and medium-sized DH areas. In other words, the excess heat utilization potentials are distributed unevenly across regions. In total, the 16 regions with the highest utilization potential account for 25% of the total utilization potential of industrial excess heat (Fig. 9).

5. Discussion

Our findings suggest that there will be a significant reduction in the industrial excess heat available in a climate-neutral energy system compared to current levels. The contribution to a climate-neutral heating sector by utilizing excess heat in DH systems is limited, even though locally this may cover a large share of DH demand. We used the process-specific method of excess heat factors multiplied with production volumes to calculate the georeferenced excess heat potential, and a spatial matching algorithm to estimate its utilization in DH areas. The excess heat potentials of low-carbon industrial processes add up to 9–50 TWh per year, which is 78%–92% lower than current levels and also lower than the figures in other studies quantifying industrial excess heat,

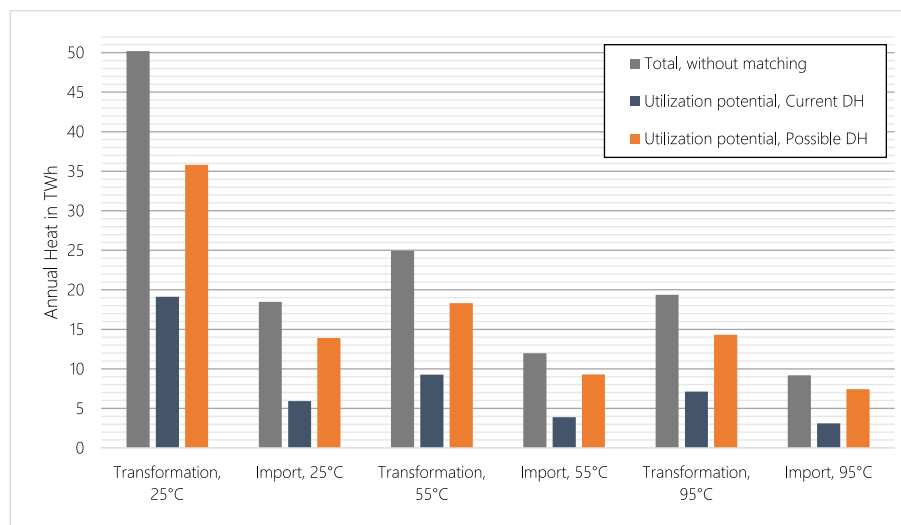


Fig. 6. Total excess heat potential from low-carbon processes available for external use and utilization potentials spatially matched with Current and Possible DH areas from Ref. [47], distinguished by temperature level in a climate-neutral energy system.

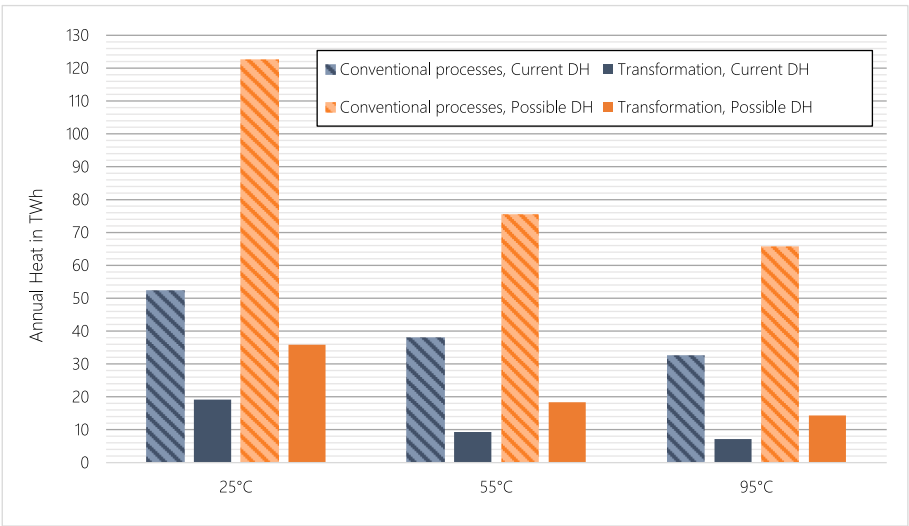


Fig. 7. Comparison of utilization potentials of excess heat for conventional and low-carbon processes, each matched with Current and Possible DH areas.

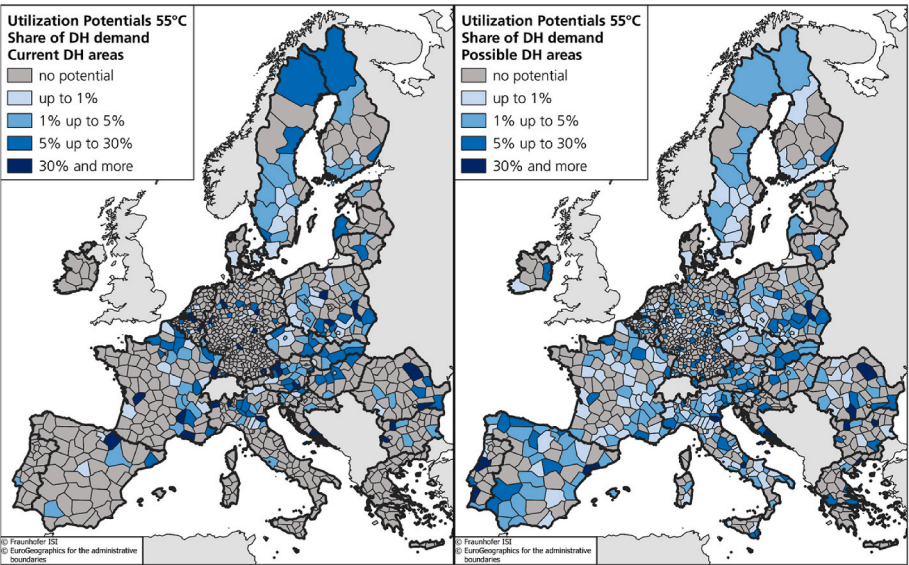


Fig. 8. Share of excess heat for DH demand per NUTS 3 region. Depicted for the transformation scenario, the temperature of 55 °C and matched with Current DH areas (left) and Possible DH areas (right).

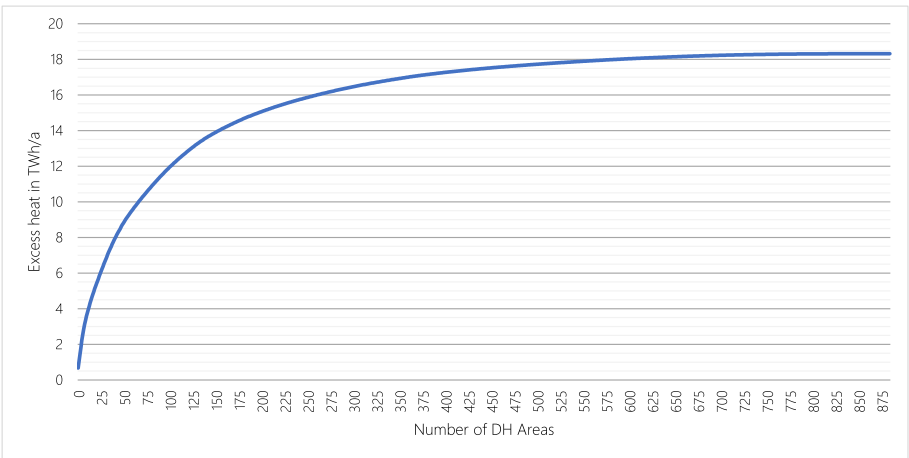


Fig. 9. Utilization potential aggregated over DH Areas. Transformation scenario, 55 °C, spatially matched with Possible DH areas.

which range from 80 to 800 TWh per year [24–27]. Process changes in industry and the exploitation of efficiency potentials are the main influencing factors, but also the general decrease in the production of basic primary materials in the EU (driven by an assumed shift towards a circular economy and secondary production routes). The excess heat available is based on ambitious assumptions regarding efficiency, internal heat utilization and the diffusion of low-carbon processes, and therefore indicates a lower limit. Matching the excess heat sources to DH areas shows that excess heat can supply up to 4% of the DH demand for a system temperature of 55 °C, which is lower than the estimations of other studies based on current industry processes [26,27,29]. Decreasing the system temperatures increases the amount of heat that can be recovered and can partially compensate the reduction of industrial excess heat potentials.

In absolute terms, the excess heat potential from energy-intensive processes that could be utilized in DH systems in a climate-neutral system in the EU is in the range of 3–36 TWh per year. Regionally, the utilization potential could be substantially higher, generally in small and medium-sized DH areas. Expanding DH from current levels to ambitious market shares could roughly double the utilization of industrial excess heat. Limiting factors not only include spatial availability, but also that the demand is too low in nearby areas. The coverage of DH systems depends on policies and regional action, as well as on the economic competitiveness of district heating. This emphasizes the need for communal heat strategies and integrated infrastructure planning [91], that take regional characteristics into account.

In general, we deal with great uncertainties in this analysis. The transformation to a climate-neutral energy system depends - among other things - on the regulatory and economic framework, available technologies and energy carriers. Regarding the transformation of the industry sector, it is only possible to estimate the excess heat of climate-neutral processes that are not yet in operation based on assumptions about their process design. We chose rather values that result in conservative numbers of future excess heat potentials. If other processes became established e.g. using hydrogen for steam generation in the pulp and paper industry instead of heat pumps or electric steam crackers instead of methanol synthesis, this could increase our quantified potentials, potentially close to current levels. The future volume of produced materials is based on the decarbonisation scenario Mix95 from the European study [9], and strongly influences the excess heat potentials. Furthermore, the future production routes of refineries are quite uncertain, especially regarding the choice of feedstock and production locations. Shifts from current locations to more favourable locations with better availability of renewable energy like electricity and hydrogen could be a factor in the future when processes are transformed. These considerations were covered in the import scenario. In this study, we assumed that all internal heat utilization potentials are exploited as a priority, as energy efficiency in industry is a key element in the transition towards climate-neutrality. We also assumed highly efficient process technologies. This is reasoned by the fact, that energy prices will increase if electrified or hydrogen-fuelled processes are established, and that newly built processes will be more efficient than already existing ones that could be more than 40 years old. Alternatively, the total excess heat from a process could be used externally, which would increase the utilization potentials for DH possibly by 40% up to 100%, depending on the temperature level [27].

Regarding the development of district heating, we considered the Current and Possible DH areas to illustrate the lower and upper limit. The 10 km maximum distance of industrial sites from DH areas is a rather generic assumption. What constitutes a suitable connection distance in the future depends mainly on the heat quantity transported, but also other factors like geographic aspects and local stakeholders. Increasing the maximum distances could increase the number of matched excess heat sources. Increasing the maximum distance from 10 km to 30 km would connect almost all excess heat sources and increase the utilization potentials by 10–50%, depending on the temperature

level and DH coverage. Furthermore, the utilization potential was estimated based on annual values. As the demand for DH in winter is by several factors higher than in summer, the contribution could be lower and utilization potentials could be further reduced by around 20–30%, depending on seasonal thermal storages [30]. Increasing the capacity of seasonal thermal storages could address the mismatch of supply and demand, but might increase the overall system costs. An additional factor that is not considered in the methodology of spatial matching is that the existence of excess heat could be a factor in establishing a new DH system to utilize the heat, even in areas with higher distribution costs. The Possible DH areas represent a high share of DH (45%) in the EU and include regions at the lower limit of economic suitability for DH, which are close to industrial excess heat sources.

A possible additional excess heat source not considered in this study are electrolyzers to produce hydrogen. Currently, these are used on small scale in the EU, but there are ambitious EU goals for domestic hydrogen production for 2030 (RePowerEU [92]). In the near future, individual electrolyzers could be larger than 100 MW capacity and studies indicate significant excess heat potentials. As a rough estimate, and given the strategic goal of establishing 40 GW electrolysis capacity in the EU by 2030 [93,94], about 150 GW of PEM-electrolyzers could be installed by 2050. With an efficiency of 92% and a temperature level of 80 °C [95], this could lead to annual excess heat potentials of 210 TWh in the EU. This value is in line with the results of EU-wide studies on the capacity and excess heat potential of electrolyzers [96,97]. These electrolyzers would be large point sources of excess heat, concentrated at several locations in the EU with high renewable potentials, which could provide heat for DH areas in their vicinity. In the future, the excess heat from electrolysis could exceed the potentials from energy-intensive industrial processes by several factors. We decided to disregard the potentials of electrolyzers for covering DH demand, as too little is known about their possible locations. Electrolyzers and their potential contribution to DH are a topic for future research.

6. Conclusion

Industrial excess heat potentials for providing heat to district heating will decrease substantially in a climate-neutral energy system. The transformation of industrial production towards low-carbon and efficient processes with internal heat utilization and a lower share of energy-intensive primary materials will reduce the amount of excess heat in the future by 78–92% compared to current levels. The possible utilization in district heating decreases accordingly by 50% compared to current levels and could supply up to 8% of the future district heating demand. The industrial excess heat potentials decrease further if energy-intensive materials are imported rather than produced at current locations. Reducing the system temperature in district heating grids and expanding the supply area could partially compensate for the decrease in industrial excess heat in the future.

In the future, the supply of heat in district heating grids needs to be based on technologies like heat pumps utilizing renewable sources like ambient heat, water resources (ground or surface) or low-temperature excess heat (e.g. waste water treatment, server cooling), as well as deep geothermal heat. Excess heat from energy-intensive industries may only cover a small share of the district heating supply. At the local level, however, industrial sites are large point sources of excess heat and can provide a considerable share of the heat for district heating. The contribution of industrial excess heat depends on the future costs of the competing heat sources. District heating expansion and densification will increase the possible utilization of industrial excess heat.

These results indicate that policies and regional action in the form of regional heat strategies are essential to increase the competitiveness of district heating even with a decreasing heating demand, as well as to identify and utilize regional excess heat potentials that could have a great contribution. The planning of district heating and the utilization of excess heat should take into account that latter could decrease in the

future, depending on which transformation pathways are taken by industry. Local heat strategies should include such uncertainties.

Several processes analysed here are still being developed or are not yet being operated on an industrial scale. Further research is needed about the process design of electrified or hydrogen-fuelled processes in order to refine the excess heat factors applied in this paper. Other research should investigate the excess heat potentials from new products needed for a climate-neutral energy system, especially the excess heat from electrolyzers. We have made conservative estimates of the available excess heat of future low-carbon processes to show the minimum contribution of industrial potentials.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Datasets related to this article can be found at <https://doi.org/10.24406/fordatis/252> (Manz, P.; Fleiter, T.; Eichhammer, W., 2023), an online data repository hosted at Fordatis - Research Data Repository of Fraunhofer-Gesellschaft.

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Annex.

Table A1

Available excess heat potentials from industrial sites per sector for external use for all scenarios and temperature levels in TWh per year.

Temperature	Scenario	Iron and Steel	Non-ferrous Metals	Pulp and Paper	Non-metallic Minerals	Refineries and Chemicals	Total for external use
25°C	Status quo, current production	29.5	2.4	23.9	55.2	120.7	231.8
	Efficiency progress, current production	16.5	1.4	20.1	26.0	110.8	174.8
	Efficiency progress, future production	5.9	1.4	18.8	21.4	80.2	127.7
	Transformation scenario	3.2	1.4	–	18.7	26.8	50.2
	Import scenario	1.5	1.4	–	15.0	0.6	18.5
55°C	Status quo, current production	23.4	2.0	8.4	41.4	40.6	115.8
	Efficiency progress, current production	10.5	1.1	5.5	15.4	33.6	66.1
	Efficiency progress, future production	4.7	1.1	5.1	12.3	24.2	47.4
	Transformation scenario	1.6	1.1	–	11.7	10.6	25.0
	Import scenario	1.2	1.1	–	9.5	0.2	12.0
95°C	Status quo, current production	21.3	1.9	6.5	37.3	29.4	96.5
	Efficiency progress, current production	8.4	1.0	3.8	12.1	22.9	48.2
	Efficiency progress, future production	3.5	1.0	3.5	9.7	15.4	33.0
	Transformation scenario	1.2	1.0	–	9.7	7.6	19.4
	Import scenario	0.9	1.0	–	7.2	0.1	9.2

Table A2

Industrial excess heat potentials for the industry scenarios and temperature levels matched with Current and Possible DH areas from Ref. [47] in TWh per year.

Scenarios	Total for external use (without matching)	Supply potential with Current DH	Supply potential with Possible DH	Utilization potential with Current DH	Utilization potential with Possible DH
Transformation 25°C	50.2	25.6	50.1	19.1	35.8
Transformation 55°C	25.0	12.7	24.9	9.3	18.3
Transformation 95°C	19.4	9.8	19.3	7.1	14.3
Import 25°C	18.5	8.4	18.4	5.9	13.9
Import 55°C	12.0	5.4	11.9	3.9	9.3
Import 95°C	9.2	4.2	9.1	3.1	7.4

Table A3

District heating demand and industrial excess heat supply and utilization potentials for the transformation scenario for DH, 55 °C, per member state, in GWh per year.

Country	Current DH Demand	Possible DH Demand	Supply potential with Current DH	Supply potential with Possible DH	Utilization potential with Current DH	Utilization potential with Possible DH
AT	9,229	25,220	695	614	304	504
BE	4,888	41,007	1,655	1,693	1,233	1,605
BG	1,773	6,264	81	195	41	106
CY	–	146	–	–	–	–
CZ	10,174	26,936	398	394	316	354
DE	67,018	344,781	2,984	5,434	2,795	4,751
DK	7,454	19,946	8	8	7	8
EE	1,588	3,829	2	3	2	3
EL	129	12,791	–	245	–	215
ES	7,545	63,998	208	2,114	208	1,716
FI	9,636	25,631	195	188	160	169
FR	48,514	177,533	1,765	3,445	1,754	2,698
HR	1,429	4,679	195	253	167	220
HU	4,629	16,319	250	270	101	183
IE	1,695	8,187	–	249	–	74
IT	22,167	171,225	691	3,666	689	2,664
LT	1,455	4,458	327	319	30	68
LU	585	3,189	22	115	22	115
LV	2,176	5,923	16	16	10	16
MT	–	332	–	–	–	–
NL	6,973	66,093	243	1,305	237	644
PL	22,242	60,318	1,697	2,128	661	1,012
PT	581	4,508	15	491	15	258
RO	4,957	14,641	384	869	174	423
SE	12,194	34,564	281	274	156	176
SI	1,139	3,150	76	79	36	41
SK	3,564	9,407	480	525	153	297
EU-27	253,734	1,155,078	12,668	24,891	9,272	18,319

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